Name of the Student Aryan Prajapati Roll No. 22

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY

DEPARTMENT OF ELECTRICAL ENGINEERING

Mid-Term Examination March - 2025

Course: **B.Tech**
Subject Name: **EMEC-II**
Time: **1:30 Hour**

Semester : **IV**
Subject Code: **EE- 222**
Max Marks: **20**

Note: **All questions are compulsory. Draw phasor diagrams wherever necessary.**

Q.No.	Question	Marks	CO's
1 (a)	Why does a single-phase motor not have starting torque but has the running torque? Explain this with reference to double field revolving theory.	02	CO4
(b)	Explain the equivalent circuit of Single-phase Induction Motor with the help of relevant diagrams.	03	CO5
2.	Analyze the impact of change in (a) Rotor resistance (b) Supply voltage on (i) Starting torque (ii) Maximum torque (iii) Slip at maximum torque for a three-phase induction motor.	05	CO1
3.	A 480 V, 60Hz, 50 hp, three-phase induction motor is drawing 60 A at 0.85 pf lagging. The stator copper losses are 2 kW and the rotor copper losses are 700 W. The friction and windage losses are 600 W, the core losses are 1800 W and the stray losses are negligible, find: a) The air gap power. b) The Mechanical power. c) The shaft power. d) The efficiency of the motor.	05	CO1
4.	A three-phase, 50Hz, four-pole, 12 kW, 400V slip-ring induction motor, with its slip-rings, short-circuited, develops rated output at rated voltage and frequency. At a slip of 5 per cent, the maximum torque occurs with zero external resistance and it is 1.5 times the full-load torque. Neglecting stator resistance and rotational losses, calculate the following a) Slip and rotor speed at full-load torque. b) Rotor ohmic loss at full-load torque. c) Starting torque at rated voltage and frequency. d) Starting current in terms of full-load current. e) Stator current in terms of full-load current at maximum torque. f) Efficiency at full-load.	05	CO